PROPWISE AI

Exploratory Data Analysis

india_housing_prices.csv · 250,000 listings · 20 analyses · generated 28 Oct 2026 12:44 UTC

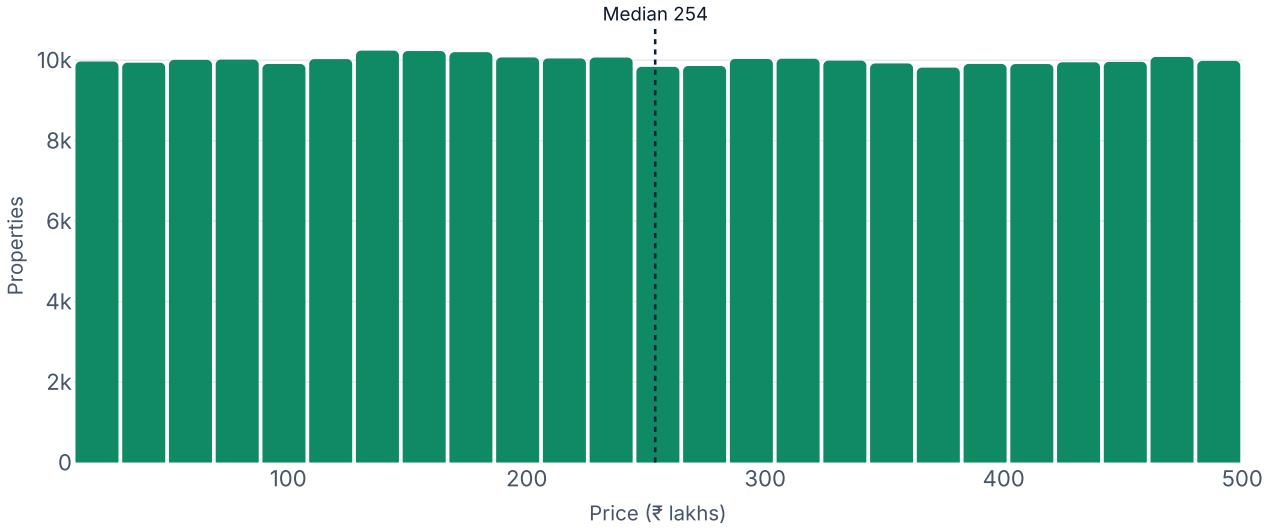
Key takeaways

- The data is clean but synthetic-looking: no missing values or duplicates, but every column behaves like an independent random draw.
- Price is unrelated to size, location, BHK, amenities, transport, or any other field (all $|r| < 0.01$). Learned valuation models cannot beat predicting the average ($R^2 \approx 0$).
- ₹/sqft varies only because price and size vary independently, so large homes look cheap per sqft by arithmetic.
- Many records are internally inconsistent (floor above building height in 46% of rows; 'under construction' homes built years ago). They are kept and flagged.
- Implication: PropWise compares a listing to its comparable cohort using a transparent rule and projects value with explicit growth scenarios. It does not claim to predict real returns.

PRICE & SIZE

1. Distribution of property prices

Distribution of asking prices



WHAT IT SHOWS

Number of listings in each ₹ lakh price band.

MAIN FINDING

Prices run from ₹10.0 L to ₹500.0 L with a median of ₹253.87 L. The histogram is flat (skew 0.008), with every band holding about the same number of listings. That is what a uniform random draw looks like, not a real market's typical right-skewed shape.

WHY IT MATTERS TO AN INVESTOR

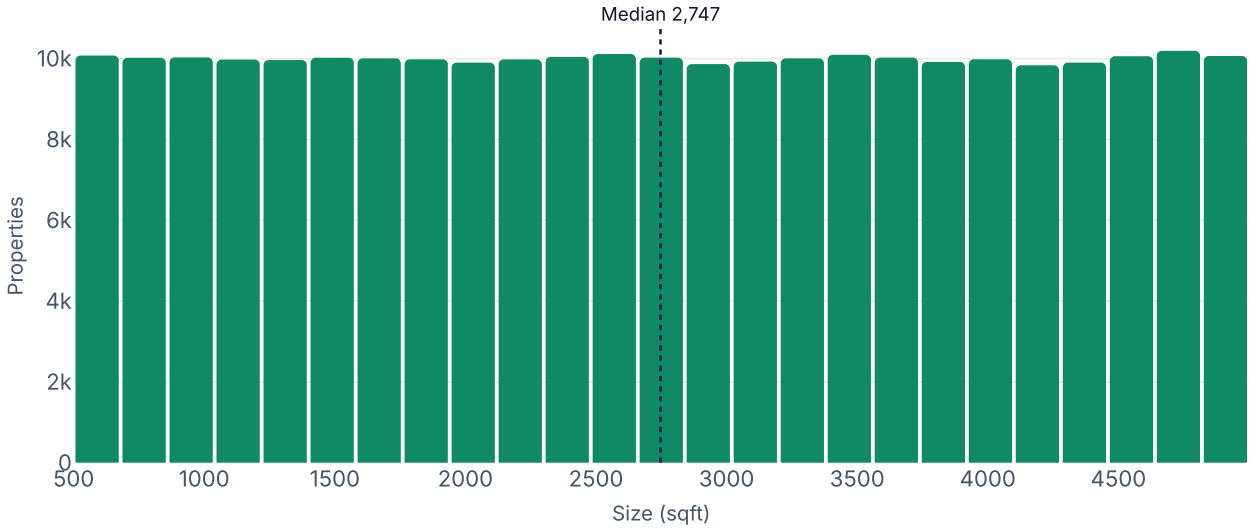
Investors usually expect many mid-market homes and a thin luxury tail. Here no price band is more 'typical' than another, so 'cheap' or 'expensive' only makes sense relative to comparable properties.

LIMITATIONS

One snapshot with no dates. Prices are asking prices, not transaction prices.

2. Distribution of property sizes

Distribution of property sizes



WHAT IT SHOWS

Number of listings in each size band (sqft).

MAIN FINDING

Sizes run from 500 to 5,000 sqft (median 2,747). The distribution is again flat (skew 0.001).

WHY IT MATTERS TO AN INVESTOR

Size is the denominator of ₹/sqft, so it shapes every value comparison in this product.

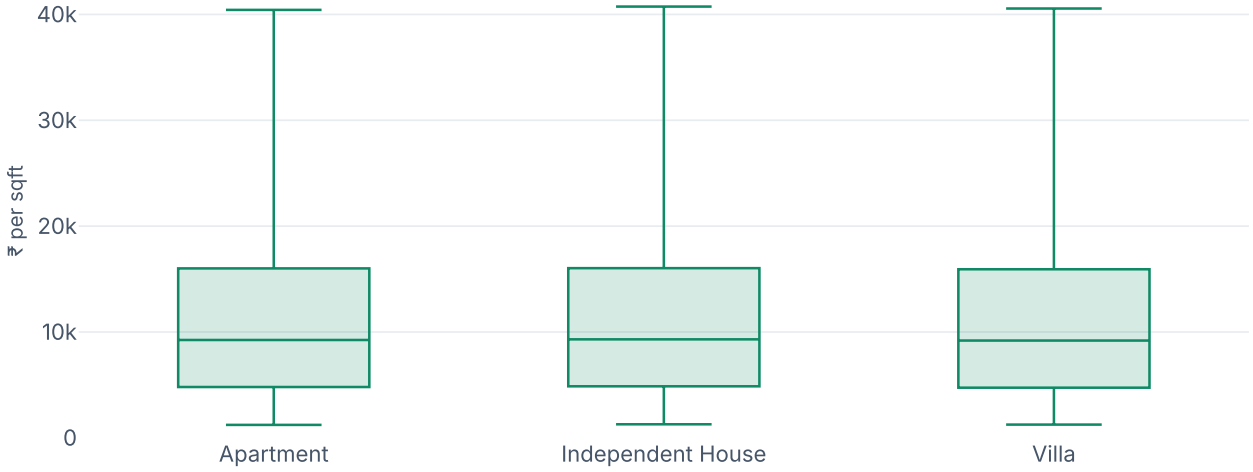
LIMITATIONS

Size is not linked to BHK: 44.5% of 1-BHK listings exceed 3,000 sqft, which is implausible.

3. Price per sqft by property type

Price per sqft by property type

Box = quartiles · whiskers = 5th–95th percentile



WHAT IT SHOWS

Spread of ₹/sqft (quartiles, 5th–95th percentile whiskers) for apartments, independent houses and villas.

MAIN FINDING

Median ₹/sqft: Apartment ₹9,245, Independent House ₹9,305, Villa ₹9,190. The three distributions are practically identical.

WHY IT MATTERS TO AN INVESTOR

In real markets villas and apartments carry different per-sqft prices. Here property type does not separate value, so comparisons still stay within type to be safe.

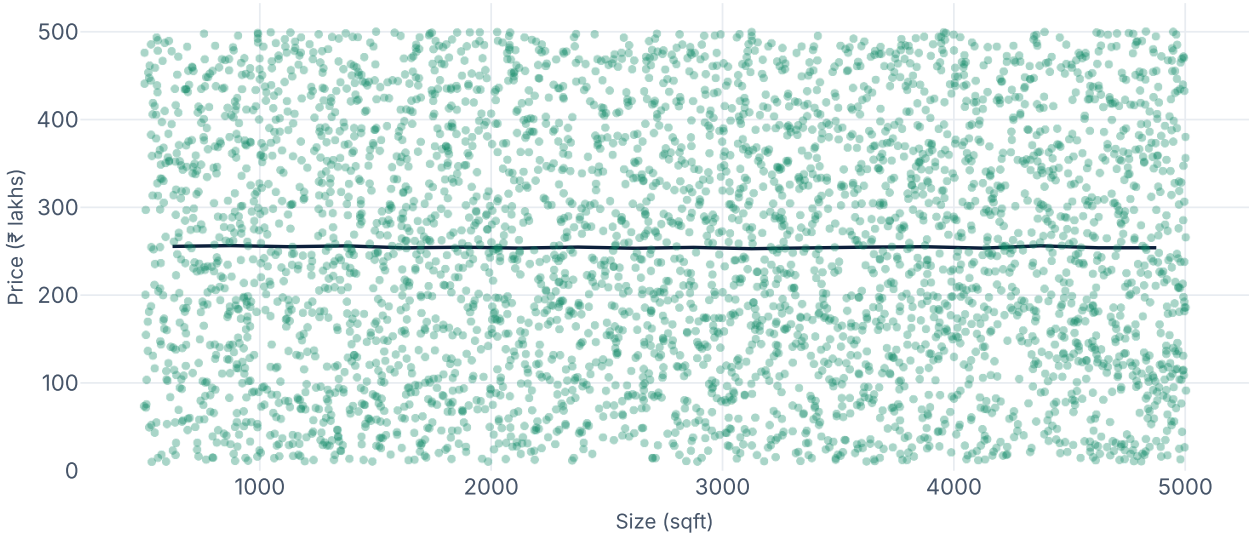
LIMITATIONS

The dataset behaves like independently generated columns, so the absence of a relationship here says nothing about real Indian housing markets.

4. Relationship between size and price

Size vs asking price

Pearson r = -0.003 · 4,000-point sample · line = binned mean



WHAT IT SHOWS

A 4,000-listing sample of size against asking price, with the binned mean price.

MAIN FINDING

Correlation is r = -0.0025, so larger homes are not more expensive. As a direct consequence, size and ₹/sqft are strongly negatively correlated (r = -0.615).

WHY IT MATTERS TO AN INVESTOR

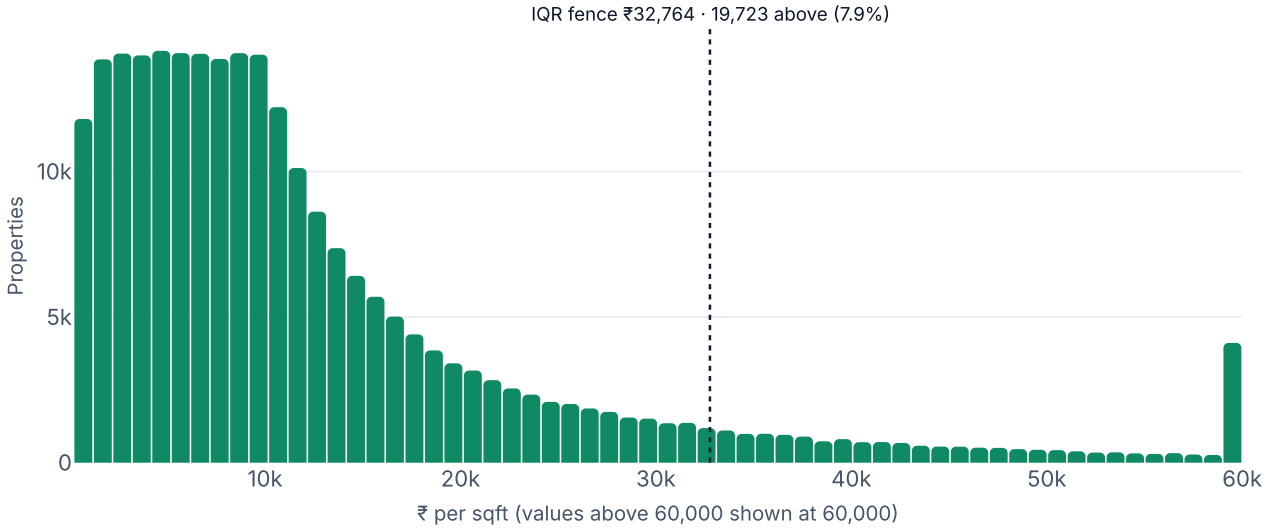
This is the single most important fact for interpreting results. Large homes look 'cheap per sqft' purely arithmetically. The good-value label compares like-for-like cohorts but cannot remove the effect inside a cohort.

LIMITATIONS

Scatter is sampled for speed. Correlation captures linear association only.

5. Outliers in price per sqft and size

Price per sqft with IQR outlier fence



WHAT IT SHOWS

Distribution of ₹/sqft with the Tukey upper fence (Q3 + 1.5×IQR).

MAIN FINDING

19,723 listings (7.9%) sit above ₹32,764/sqft. 74% of them are under 1,000 sqft, so they are small homes, not data-entry errors. Size has no outliers (bounded 500–5,000).

WHY IT MATTERS TO AN INVESTOR

Outliers are kept because they come from the same generating process as the rest. Tree models and median-based benchmarks are robust to them.

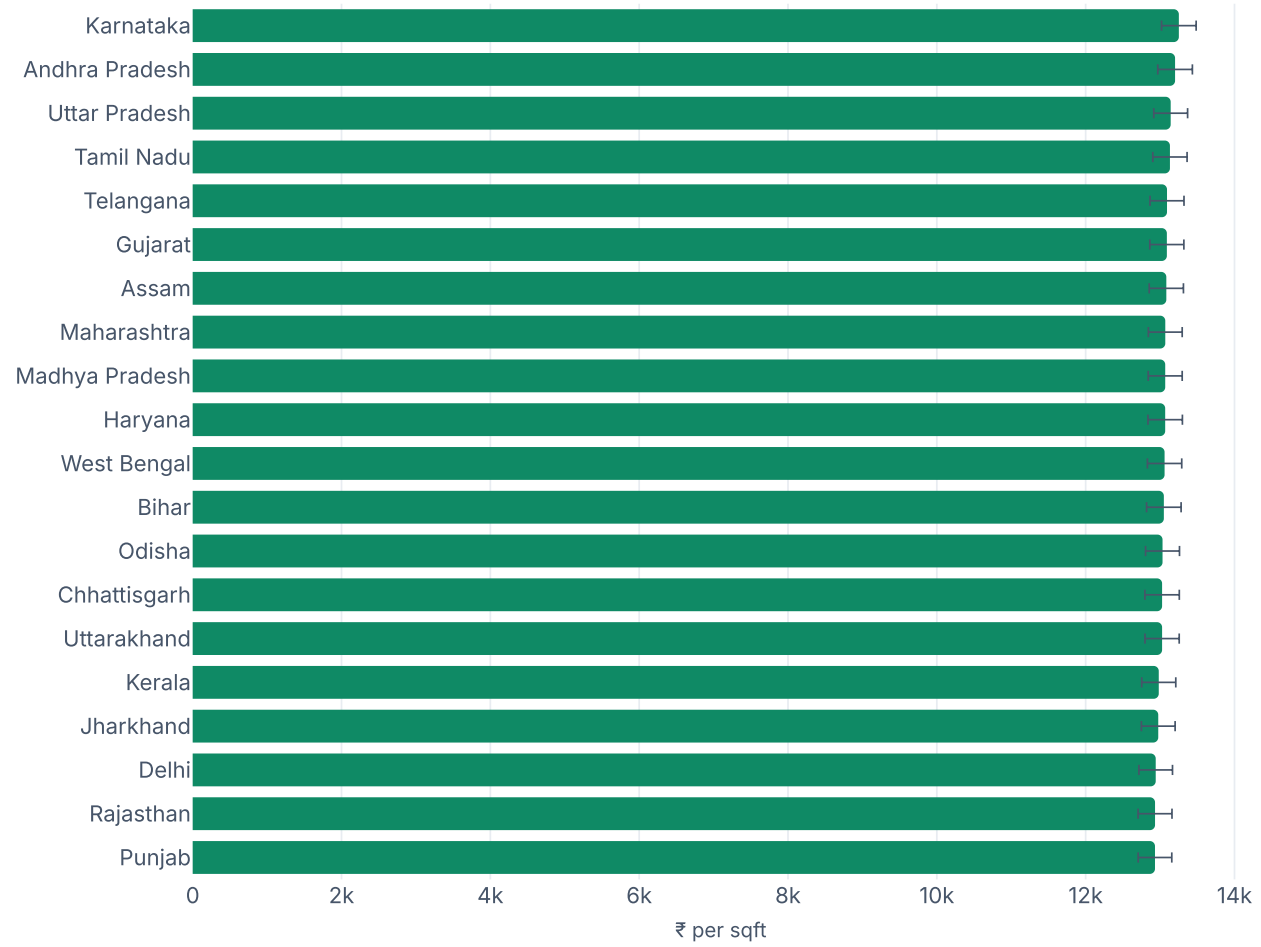
LIMITATIONS

The fence is a convention, not a rule for 'wrong' data.

6. Average price per sqft by state

Average ₹/sqft by state

Range across groups: 2.5% · bars = mean ± 95% CI



WHAT IT SHOWS

Mean ₹/sqft per state with 95% confidence intervals.

MAIN FINDING

Highest: Karnataka; lowest: Punjab. The gap is only 2.5% and the confidence intervals overlap.

WHY IT MATTERS TO AN INVESTOR

Real state-level differences (e.g. Maharashtra vs. Bihar) are large. Their absence confirms that location carries no price signal in this data.

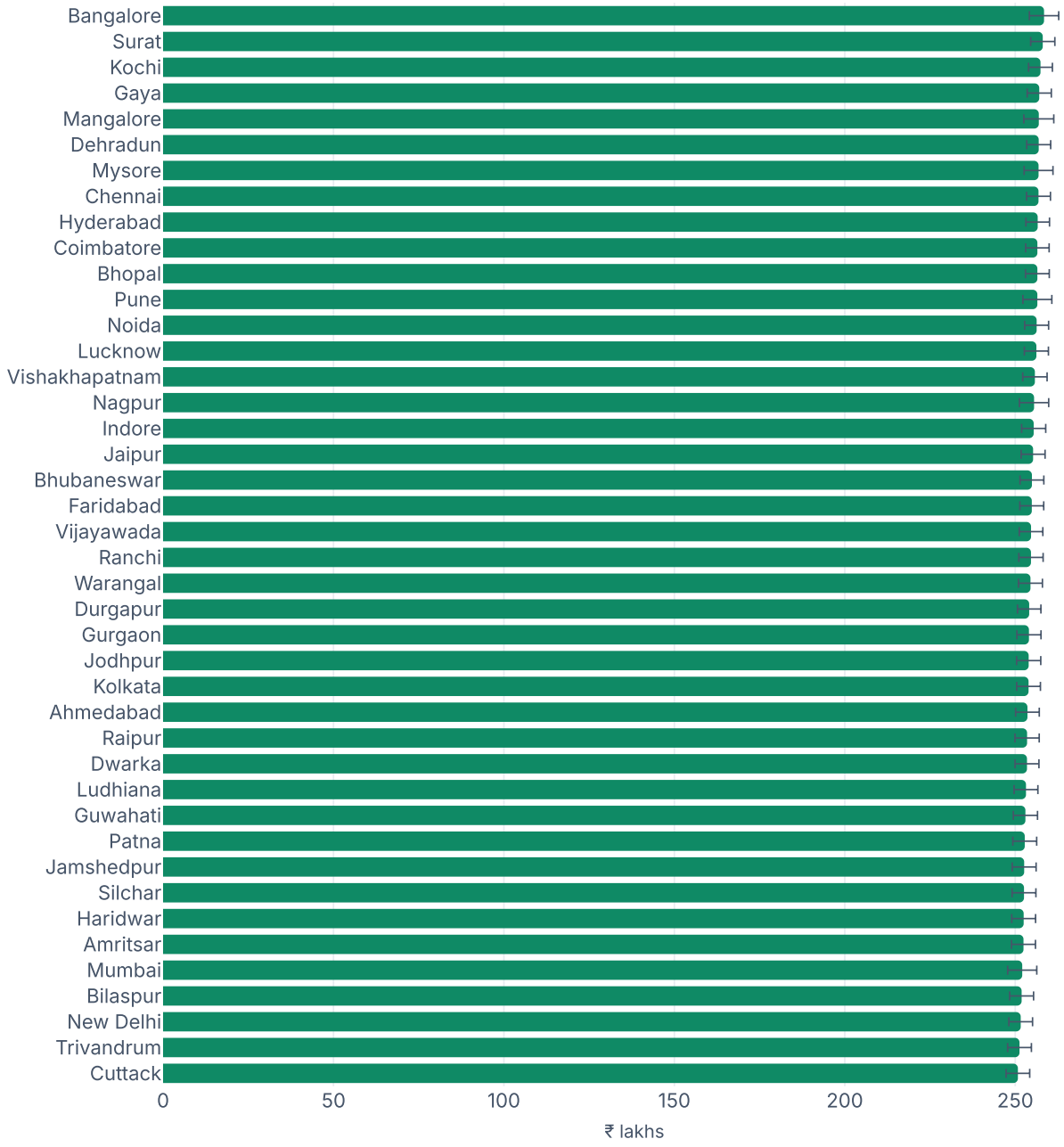
LIMITATIONS

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7. Average property price by city

Average asking price by city

Range across groups: 3.1% · bars = mean ± 95% CI



WHAT IT SHOWS

Mean asking price per city with 95% confidence intervals.

MAIN FINDING

Bangalore is highest (₹258.5 L) and Cuttack lowest (₹250.8 L): a 3.1% spread, within sampling noise.

WHY IT MATTERS TO AN INVESTOR

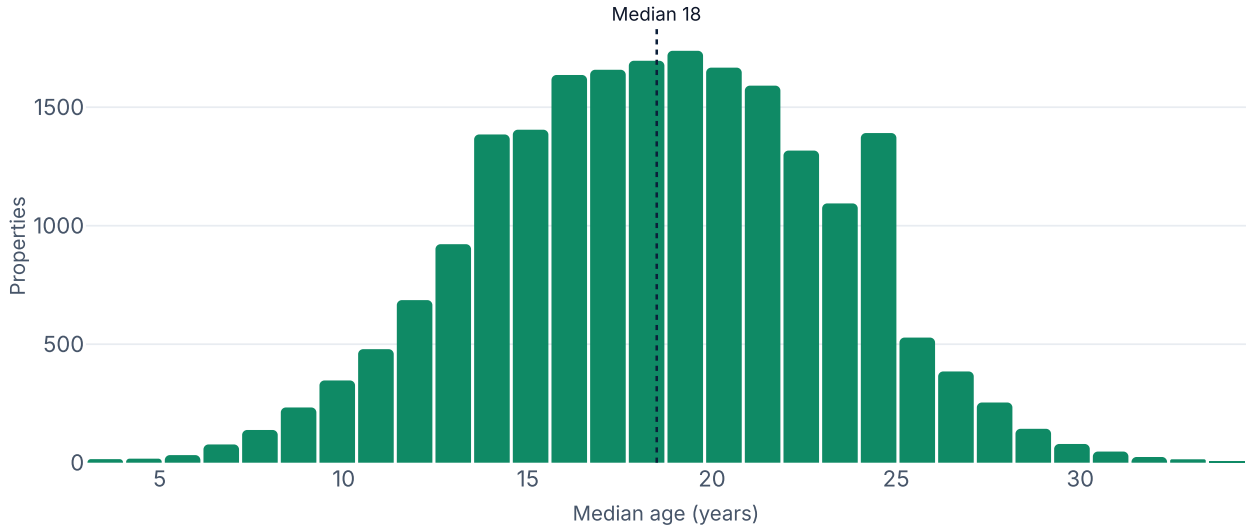
Mumbai and Gaya at the same average price is not realistic. Use city benchmarks only as within-dataset references.

LIMITATIONS

The dataset behaves like independently generated columns, so the absence of a relationship here says nothing about real Indian housing markets.

8. Median property age by locality

Median property age across City–Locality pairs



WHAT IT SHOWS

Distribution of median building age across the 20,999 City–Locality pairs (locality IDs repeat in every city, so the pair is the real unit).

WHY IT MATTERS TO AN INVESTOR

Locality-level statistics would be over-fitted, so the models use city-level cohorts instead.

MAIN FINDING

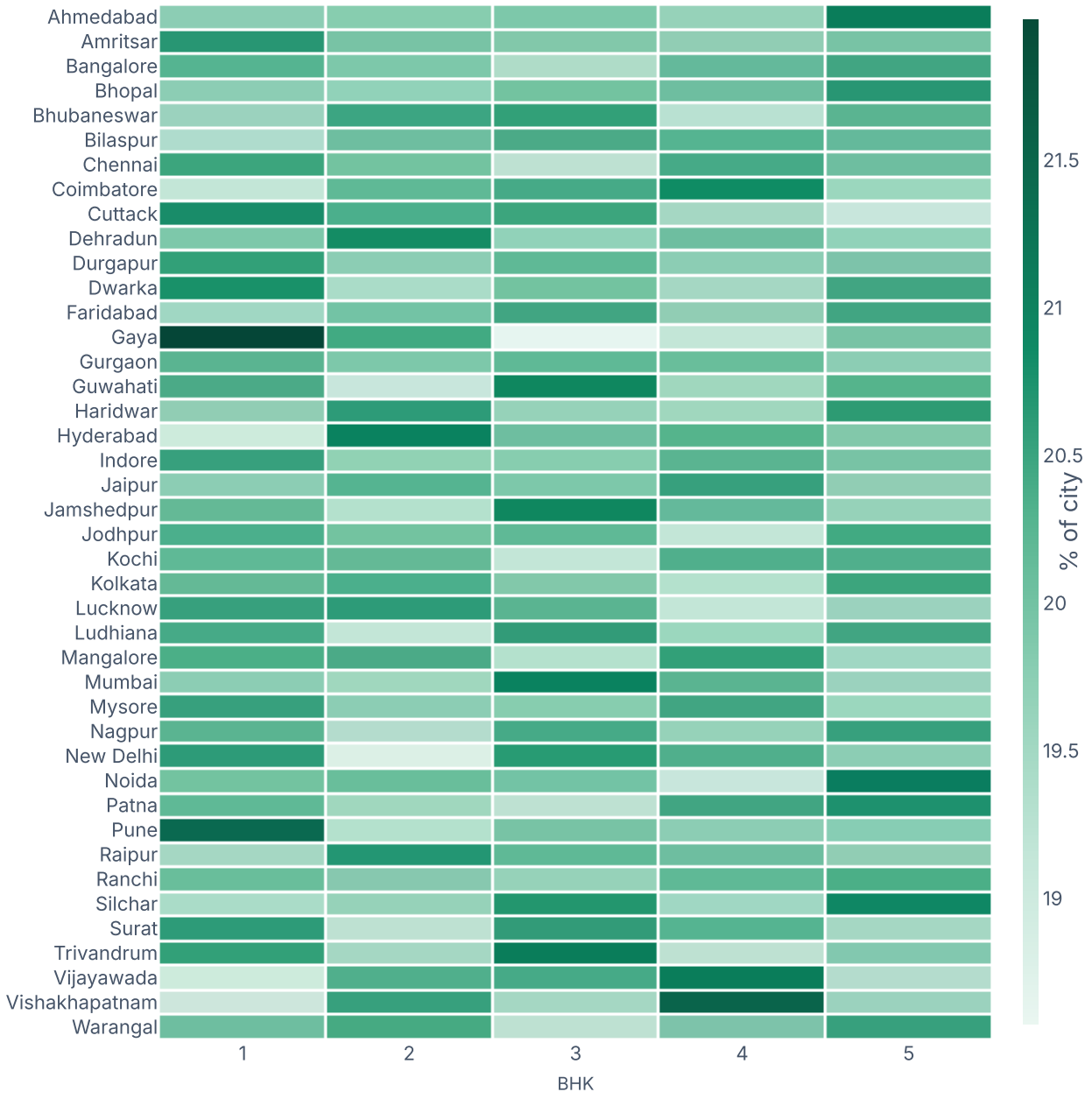
Half of localities have a median age between 15 and 22 years. Each locality has only about 12 listings, so locality medians are noisy.

LIMITATIONS

Small samples per locality, and locality IDs are anonymised.

9. BHK distribution across cities

BHK mix by city (% of each city's listings)



WHAT IT SHOWS

Share of each city's listings by BHK.

MAIN FINDING

Every cell lies between 18.6% and 22.0%: each city has a near-even 20% split across 1–5 BHK.

WHY IT MATTERS TO AN INVESTOR

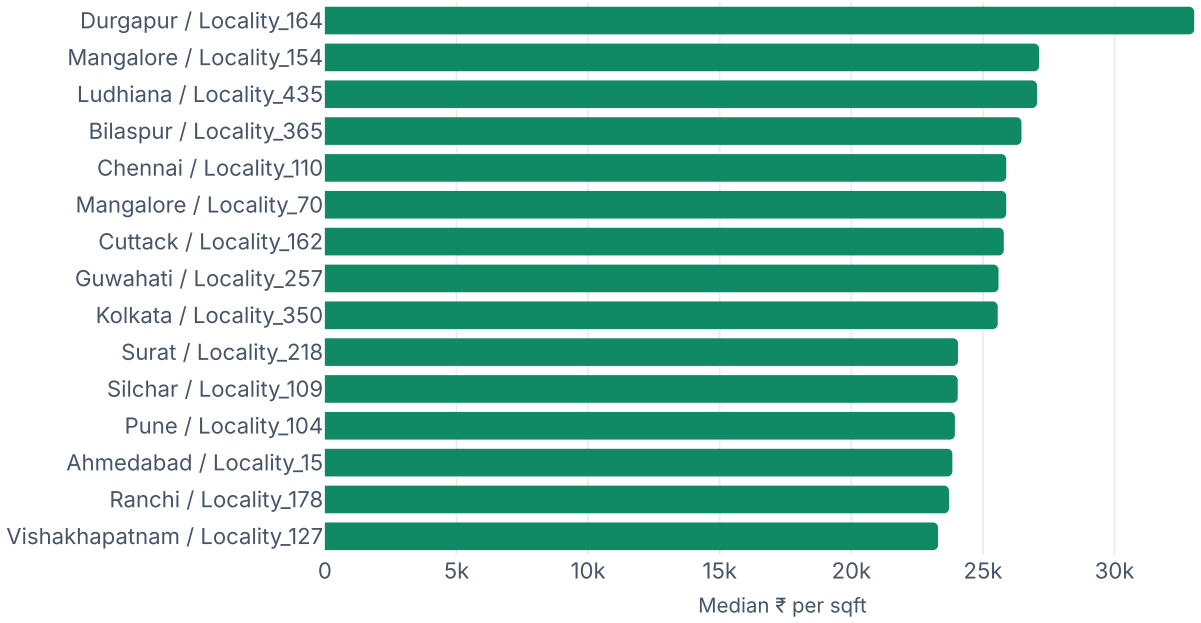
Unit mix normally differs by city (more 1–2 BHK in metros). An even mix means cohorts are well balanced, at least 165 training listings each.

LIMITATIONS

The dataset behaves like independently generated columns, so the absence of a relationship here says nothing about real Indian housing markets.

10. Most expensive localities

Highest median ₹/sqft localities (≥10 listings)



WHAT IT SHOWS

The 15 City–Locality pairs with the highest median ₹/sqft (pairs with ≥10 listings).

MAIN FINDING

Top: Durgapur / Locality_164 at ₹33,025/sqft. Locality medians have a standard deviation of ₹2,775. If prices are shuffled randomly across listings, it is ₹2,756, so locality differences are what random assignment alone produces.

WHY IT MATTERS TO AN INVESTOR

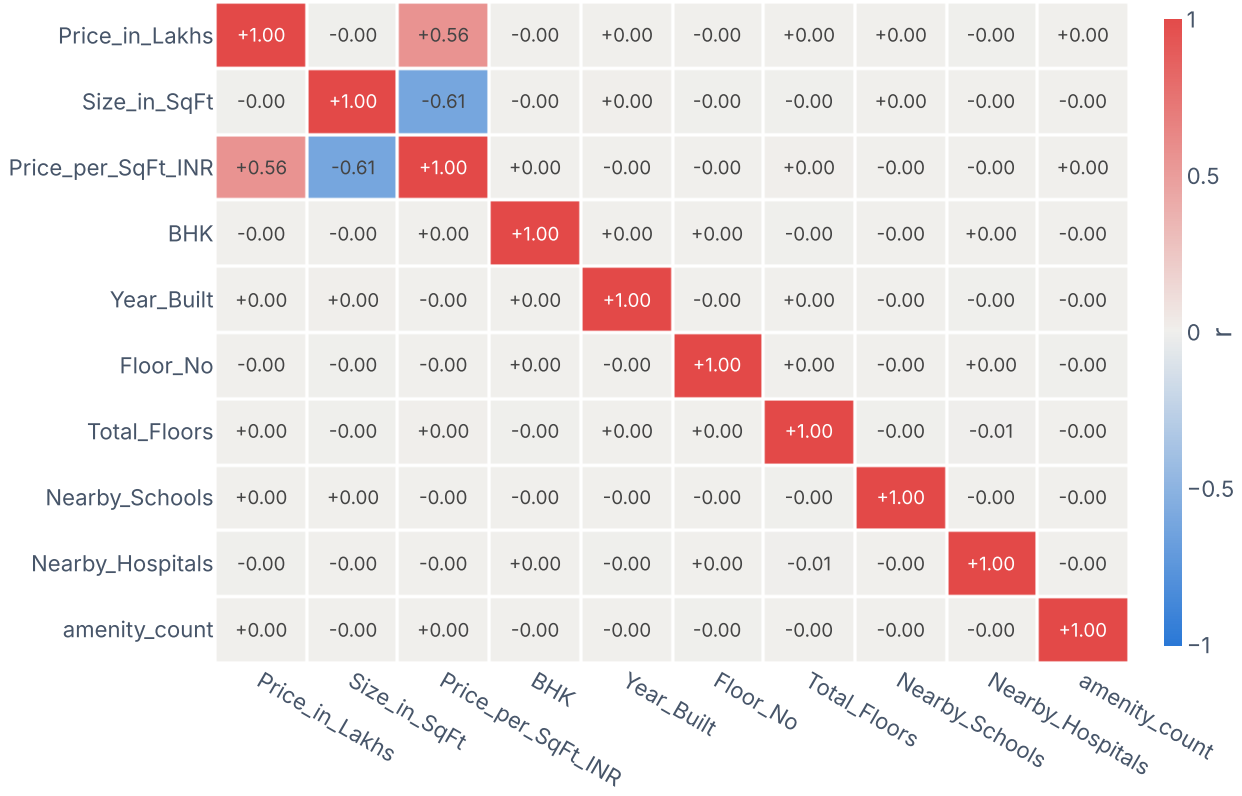
'Top localities' here are mostly small-sample luck and would likely regress to the mean. Do not read them as premium neighbourhoods.

LIMITATIONS

The brief asked for price trends of top localities, but there is no time dimension, so trends cannot be computed.

11. Correlation among numeric variables

Correlation among numeric variables (Pearson r)



WHAT IT SHOWS

Pearson correlation for every pair of numeric fields.

MAIN FINDING

Only the two mechanical relationships stand out: Price_per_SqFt_INR ~ Size_in_SqFt r = -0.61; Price_in_Lakhs ~ Price_per_SqFt_INR r = +0.56. Every other pair has $|r| < 0.01$ (Year Built and Age, r = −1 by definition, is excluded).

WHY IT MATTERS TO AN INVESTOR

No attribute moves with price, which is why the regression signal test finds $R^2 \approx 0$.

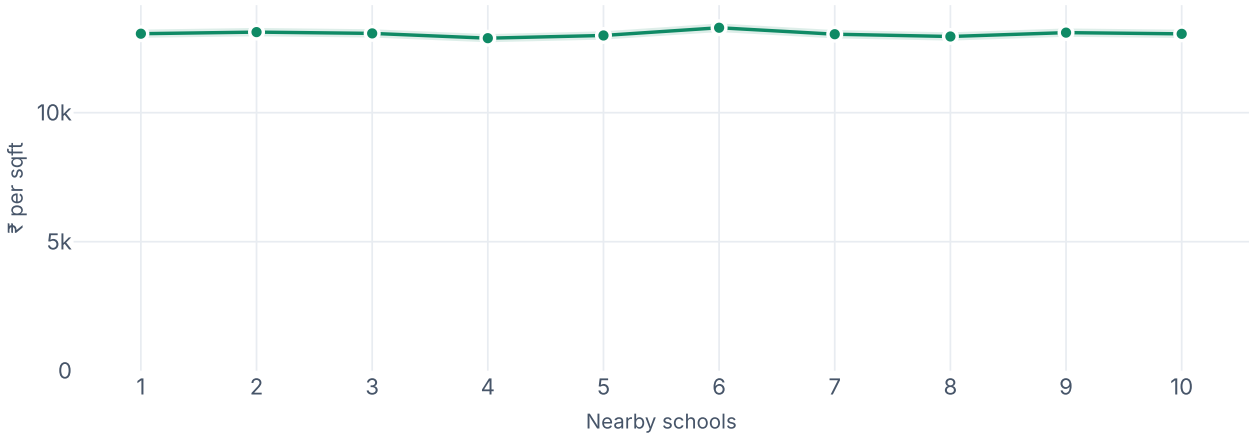
LIMITATIONS

Correlation is linear and pairwise. The models test non-linear and interaction effects separately (still $R^2 \approx 0$).

12. Nearby schools and price per sqft

Mean ₹/sqft by nearby-school count

r = -0.000 · band = 95% CI



WHAT IT SHOWS

Mean ₹/sqft at each school count (1–10) with a 95% CI band.

MAIN FINDING

Flat line (r = -0.0001).

WHY IT MATTERS TO AN INVESTOR

School access normally commands a premium. Here it doesn't, so the app shows it as context only.

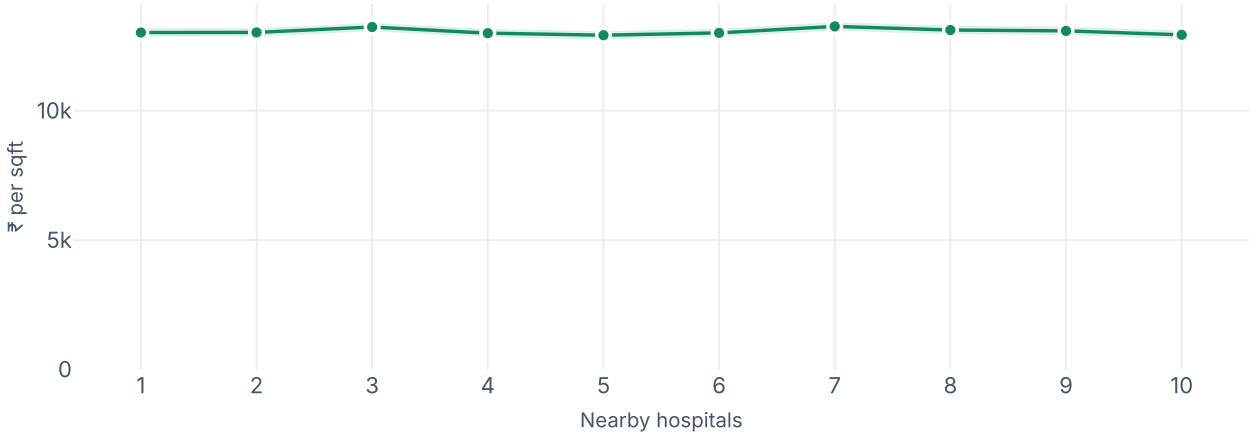
LIMITATIONS

The field may be a count or a rating; the source does not say.

13. Nearby hospitals and price per sqft

Mean ₹/sqft by nearby-hospital count

r = -0.000 · band = 95% CI



WHAT IT SHOWS

Mean ₹/sqft at each hospital count with a 95% CI band.

MAIN FINDING

Flat line (r = -0.0001).

WHY IT MATTERS TO AN INVESTOR

Same as schools: not a value driver in this dataset.

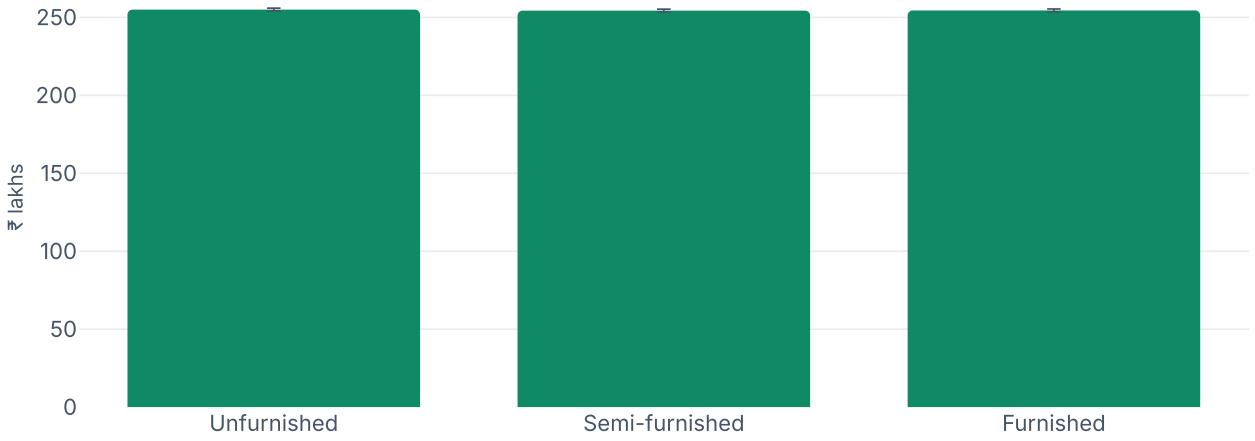
LIMITATIONS

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14. Price by furnishing status

Average price by furnishing status

Range across groups: 0.3% · bars = mean ± 95% CI



WHAT IT SHOWS

Mean asking price by furnishing level with 95% CI.

WHAT IT MATTERS TO AN INVESTOR

Furnishing normally adds a modest premium. Not here.

MAIN FINDING

Groups differ by only 0.25%: Furnished ₹254.4 L, Semi-furnished ₹254.3 L, Unfurnished ₹255.0 L.

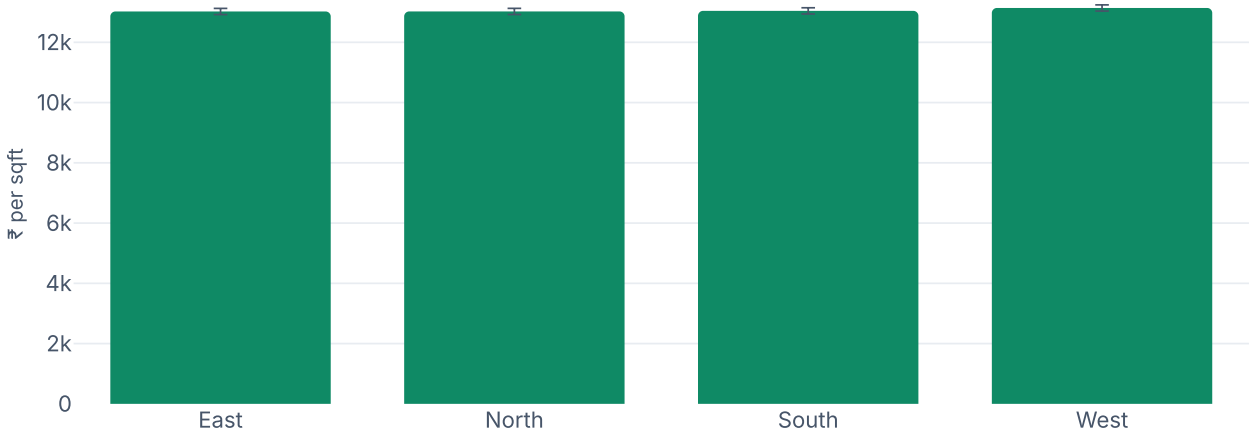
LIMITATIONS

The dataset behaves like independently generated columns, so the absence of a relationship here says nothing about real Indian housing markets.

15. Price per sqft by facing direction

Average ₹/sqft by facing

Range across groups: 0.9% · bars = mean ± 95% CI



WHAT IT SHOWS

Mean ₹/sqft by facing direction with 95% CI.

WHAT IT MATTERS TO AN INVESTOR

Vastu preferences (often east/north) can move real prices. That effect is absent here.

MAIN FINDING

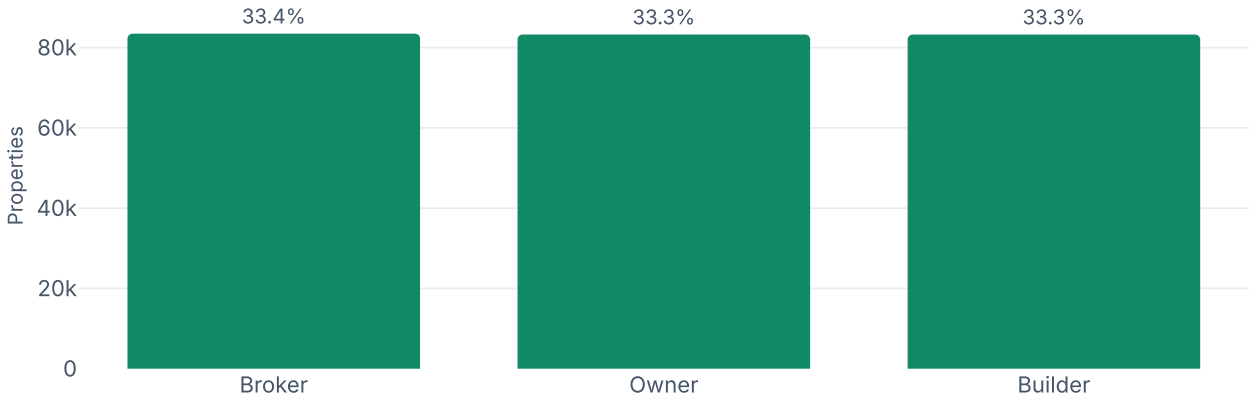
Range 0.89%. No direction is priced differently.

LIMITATIONS

The dataset behaves like independently generated columns, so the absence of a relationship here says nothing about real Indian housing markets.

16. Listings by owner type

Listings by owner type



WHAT IT SHOWS

Number of listings per seller type.

MAIN FINDING

Even split: Broker 33.4%, Owner 33.3%, Builder 33.3%.

WHY IT MATTERS TO AN INVESTOR

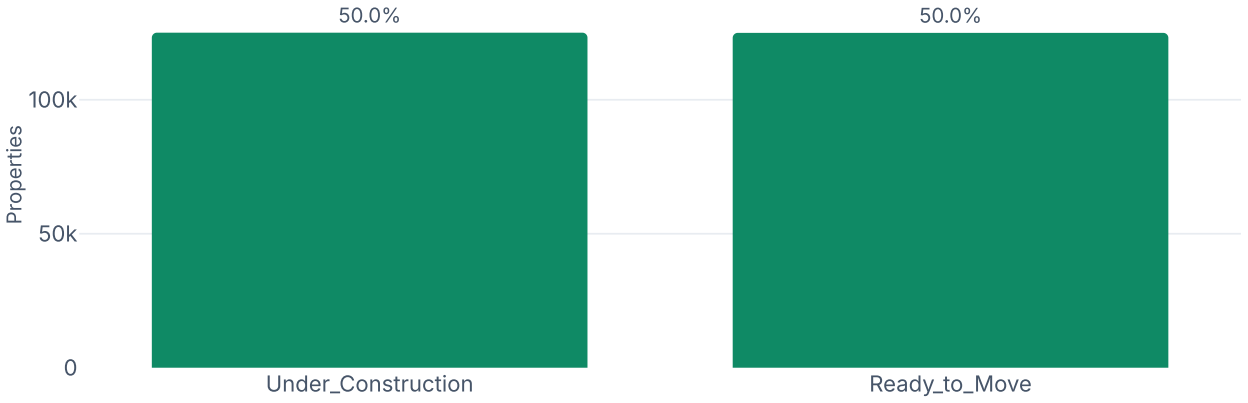
Builder vs. owner listings can differ in negotiability and fees. Brokerage is not modelled.

LIMITATIONS

Seller type only. No fees or commissions are in the data.

17. Listings by availability status

Listings by availability status



WHAT IT SHOWS

Ready-to-move vs. under-construction listings.

MAIN FINDING

Even split: Under Construction 50.0%, Ready to Move 50.0%. 91.1% of under-construction listings were built in 2020 or earlier, which is internally inconsistent.

WHY IT MATTERS TO AN INVESTOR

Under-construction purchases carry delivery risk. The app flags this as context, not as part of the verdict.

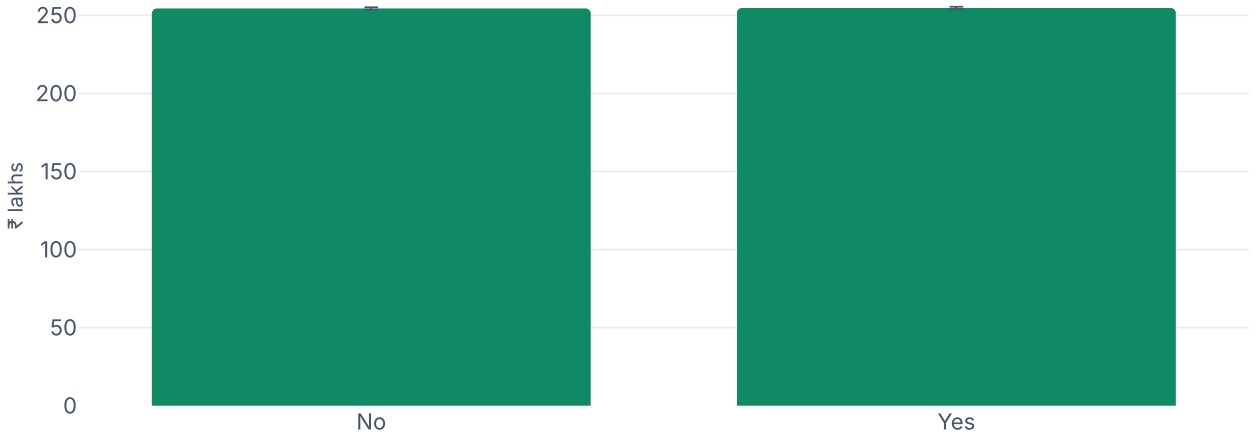
LIMITATIONS

Inconsistent records are kept and flagged, not deleted.

18. Parking and price

Average price by parking availability

Range across groups: 0.1% · bars = mean ± 95% CI



WHAT IT SHOWS

Mean asking price with and without parking.

MAIN FINDING

Difference: 0.12%.

WHY IT MATTERS TO AN INVESTOR

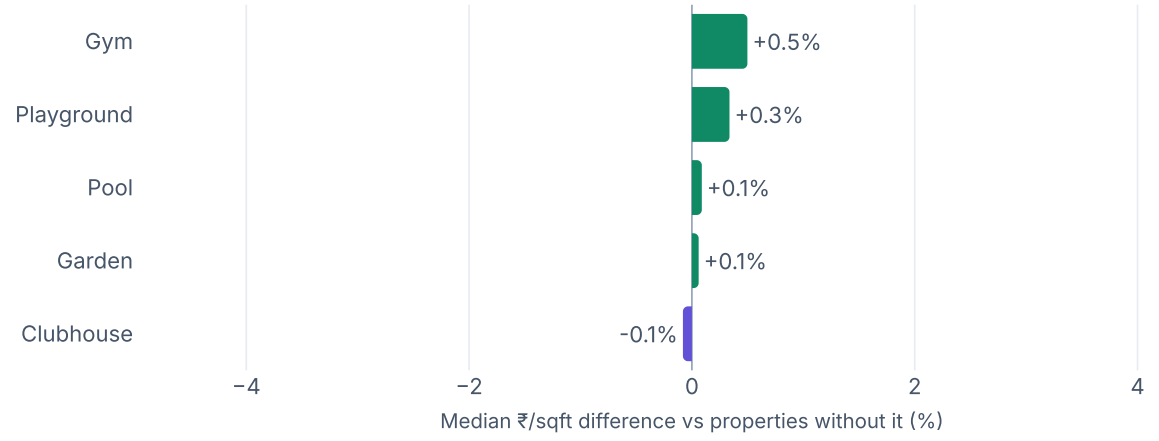
Parking would usually add value in dense cities. No measurable effect here.

LIMITATIONS

Yes/no only. The number of spaces is not recorded.

19. Amenities and price per sqft

Price per sqft premium by individual amenity



WHAT IT SHOWS

Median ₹/sqft difference for listings with each amenity vs. without it.

MAIN FINDING

Every amenity moves the median by about ±1%. Across 1–5 amenities, mean ₹/sqft varies by only 0.67%.

WHY IT MATTERS TO AN INVESTOR

Amenity count is shown as context in the analyzer, not as a value driver.

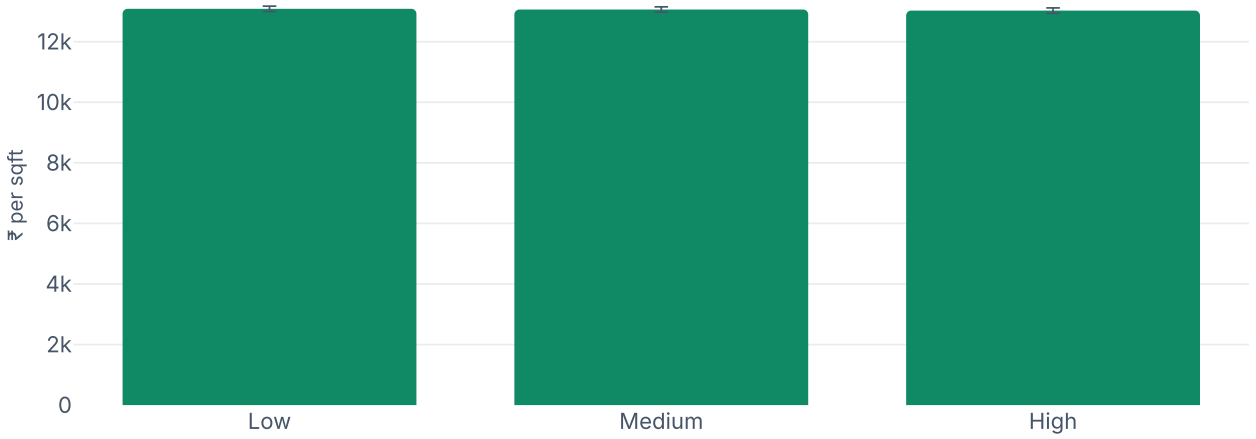
LIMITATIONS

Amenity quality and maintenance charges are not recorded.

20. Public transport accessibility and price per sqft

Average ₹/sqft by transport access

Range across groups: 0.4% · bars = mean ± 95% CI



WHAT IT SHOWS

Mean ₹/sqft by transport accessibility level.

WHY IT MATTERS TO AN INVESTOR

Metro proximity is one of the strongest real price drivers in Indian cities. Its absence is another sign that the data is synthetic.

MAIN FINDING

Range 0.45%. High-access listings are not priced higher.

LIMITATIONS

The dataset behaves like independently generated columns, so the absence of a relationship here says nothing about real Indian housing markets.

Correlation is not causation. Findings describe this dataset only and are not investment advice.